Get Started with AI in Azure

Azure provides a secure, scalable platform to build, deploy, and manage AI applications using modern tools and services — from Microsoft Foundry to AI endpoints and integrated Azure infrastructure.

PREREQUISITE: BASIC AZURE PORTAL KNOWLEDGE

Introduction to AI in Azure

AI systems perform tasks requiring human intelligence — reasoning, learning, and decision-making. AI applications are **model-driven and dynamic**, adapting and improving over time by processing text, images, or speech through model inference.

Healthcare

Disease detection and diagnostics

Finance

Fraud detection and risk analysis

Retail

Personalized recommendations

Manufacturing

Predictive maintenance

Education

Personalized learning experiences

AI applications also require infrastructure — security, networking, and storage. Azure provides all these components, enabling developers to build, deploy, and scale AI solutions efficiently.

Understand Azure

Microsoft Azure is a leading cloud platform for building and managing applications over the internet — no physical hardware required. It provides scalable, secure infrastructure for modern AI solutions.

Core Service Areas

Compute & App Services

Run and host applications without managing servers

Storage

Store and manage files, databases, and backups

Networking

Connect resources securely and enable communication

Resource Hierarchy

The **Azure Portal** is a web-based interface to create, configure, and monitor resources. Azure also supports automation via **CLI and PowerShell** for consistent, large-scale deployments.

Developing AI Apps on Azure

Azure provides everything needed to build secure, scalable AI applications — from identity management to hosting and data storage.

Security & Networking

Azure Entra ID manages identity access. Azure Key Vault protects API keys and secrets. Networking ensures safe, reliable communication between services.

Hosting & Scaling

Deploy on VMs or containers using Azure Kubernetes Service or App Service. Scale horizontally (add instances) or vertically (increase resources).

Data Storage

Azure SQL Database, Cosmos DB, and PostgreSQL handle training data, inputs, outputs, logs, and configuration.

AI Capabilities

Microsoft Foundry enables secure, scalable AI development with built-in models, agents, and tools for vision, speech, and language.

Microsoft Foundry for AI

Microsoft Foundry is a unified platform on Azure for building, deploying, and managing AI applications and agents with enterprise-level security and scalability.

Development Workflow

01

Create a Foundry Project

Set up your workspace for agents, datasets, and AI workflows

02

Select & Deploy a Model

Choose from available AI models and deploy to your resource

03

Test in the Playground

Validate prompts and outputs before integrating

04

Integrate with Your App

Connect client applications to the model via endpoints

Models

Access thousands of AI models from OpenAI, Meta, Anthropic, and more

Agents

Create agents that reason, automate tasks, and interact with data

Tools

Built-in AI services for vision, speech, and language processing

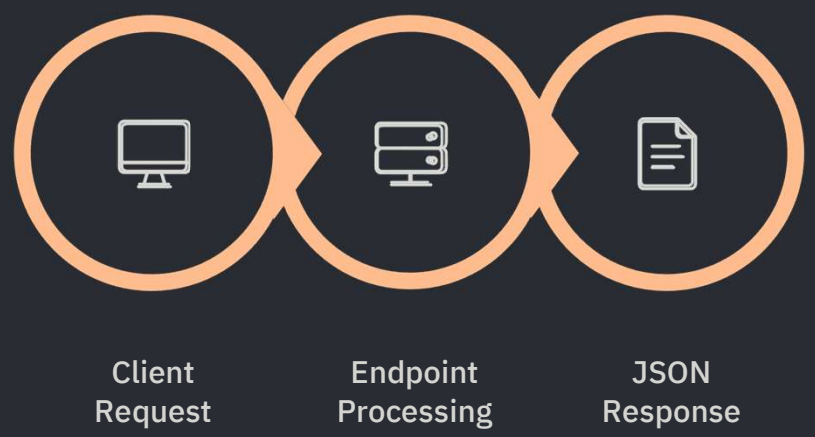
Knowledge

Foundry IQ connects and retrieves data from multiple sources securely

Using Microsoft Foundry Endpoints

Foundry resources are accessed through APIs. An **endpoint** is a unique URL that acts as an entry point to a model or service — clients send requests and receive AI-generated responses.

How It Works



Endpoint Types

Development Options

- Use **REST APIs** directly for full control
- Use **SDKs** (Python, JavaScript, C#) to simplify integration

Endpoints are protected and require authentication via API keys or identity tokens through Azure Entra ID.

Project Endpoints

Manage project-level resources, agents, and workflows

Model Endpoints

Send prompts and receive AI-generated responses directly

Module Summary

You now have the foundation to start building AI apps, deploying models, and creating intelligent agents on Azure.

Azure Infrastructure

Scalable, secure cloud platform for modern AI applications

Microsoft Foundry

Unified platform with ready-to-use models, tools, and agent capabilities

Fast Development

Build secure, scalable AI apps quickly with built-in governance

Seamless Integration

Integrate speech, vision, and language features via REST APIs or SDKs

Next Steps: Explore the Azure Portal, create a Foundry project, and experiment with model endpoints using the playground.

Key Capabilities Enabled by Foundry Endpoints

Microsoft Foundry endpoints unlock a wide range of AI capabilities, allowing developers to integrate advanced intelligence into their applications with ease and security.

Advanced Natural Language Processing

Leverage state-of-the-art language models for text generation, summarization, sentiment analysis, and multilingual translation to enhance user interaction and content creation.

Powerful Computer Vision

Integrate image and video analysis for object detection, facial recognition, scene understanding, and optical character recognition (OCR) in various applications.

Seamless Speech & Audio Integration

Implement accurate speech-to-text transcription, natural-sounding text-to-speech synthesis, and voice command processing for intuitive user interfaces.

Custom AI Model Deployment

Deploy and manage your own custom-trained machine learning models at scale, benefiting from Foundry's robust infrastructure, monitoring, and governance features.